

Making Premium Cooking Legible: A Panorama-Display Lightwave Oven

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ABSTRACT: Premium kitchen appliances are often presented through finish, brand language, and feature count. During cooking, however, their value also depends on whether the cook can understand what the appliance is doing, judge a suggested setting, and anticipate the next step. This report presents the Panorama-Display Lightwave Oven, a redesign of the LG Dios Objet Collection Lightwave Oven that treats process legibility as a central part of premium experience. The concept combines a full-width transparent Panorama Display with a single physical dial, contextual digital buttons, a handle-edge indicator, and a modular tray direction. Five AI cooking-support modules—Voice Chef, Auto Cook Agent, Chef Persona, Vision Feedback, and Taste Memory—use the same surface so that sensing, proposals, progress, and learning remain visible and editable rather than hidden inside automation. Two connected studies informed the design. User Study 1 combined six in-depth interviews, three first-time cooking sessions with concurrent think-aloud, mind mapping, rapid prototyping, flow analysis, and a hybrid affinity diagram built from more than 400 utterance-level notes in 31 clusters. It located the experience gap in three legibility problems rather than in missing features. User Study 2 compared the developed direction with a compact-display design through a Google Forms survey with 36 respondents. The developed direction was preferred on all eleven premium-experience attributes. Respondents selected it on a mean of 6.7 of these attributes, compared with 2.1 for the comparison design, and 29 of 35 respondents selected it on more premium attributes than the comparison design (exact sign test, $p < .001$). The comparison design retained advantages in first-time ease and expected everyday use, while overall experience and kitchen convenience were evenly split and the eleven-point purchase-intent ratings did not differ significantly. The results support a display-first approach to domestic AI: cooking support becomes more convincing when its state, suggestions, and progress can be read, questioned, and adjusted during use.

KEYWORDS: panorama display, cooking experience, smart appliance, human-AI interaction, concept evaluation

1. Introduction

A lightwave oven combines several kitchen tools in one cabinet. The LG Dios Objet Collection Lightwave Oven (model MLJ32HWS) includes microwave, convection, air-fryer, steam, and air sous-vide modes, connects to the ThinQ app, and follows the restrained material language of the Objet line (LG Electronics, 2024). These specifications position it as a premium product. During use, however, the same range of functions creates a series of practical decisions: which mode suits the dish, which tray level is correct, whether the suggested time can be trusted, when the door should be opened, and how a successful result can be repeated. The exterior finish does not help the cook answer these questions.

Our formative research located the gap in these everyday decisions. Participants liked the appearance and respected the brand, yet they hesitated at the dial, re-read the manual during tasks, checked the tray more than once, and made little use of automatic cooking or ThinQ. The product did not lack functions; it lacked clear ways to read the cooking process, including the machine's current state, the basis for its



Figure 1. The Panorama-Display Lightwave Oven. The front glass becomes the primary interaction surface; the handle-edge indicator keeps guidance available while the door is open.

guidance, and the next safe action. We therefore framed the problem as a mismatch between premium specifications and an experience in which the process remained difficult to interpret.

The Panorama-Display Lightwave Oven responds by starting with the interaction surface. A full-width transparent Panorama Display extends across the oven window so that cooking mode, recipe, sensed weight, remaining time, stage progress, warnings, and suggested actions can be read together over the food itself. A single physical dial and contextual digital buttons keep control direct, a handle-edge indicator continues guidance while the door is open, and a modular tray direction addresses the physical-setup problems found in the field work. Five AI cooking-support modules—Voice Chef, Auto Cook Agent, Chef Persona, Vision Feedback, and Taste Memory—use this shared surface. The display and AI support are designed as one system: the surface makes automated guidance visible and editable, while the guidance gives the wider display a clear purpose throughout cooking.

This report makes four contributions.

- **A diagnosis of premium cooking as a legibility problem.** Field evidence from interviews, first-time cooking tasks, affinity synthesis, and flow analysis shows how a feature-rich appliance can still feel effortful when operation hierarchy, physical setup, and smart-cooking feedback stay opaque.
- **A display-first product concept.** The Panorama-Display Lightwave Oven integrates the Panorama Display, a single dial with contextual digital buttons, a handle indicator, and a modular tray direction into one interaction system, with the display carrying state, progress, and physical-action cues.
- **An inspectable AI cooking layer.** Five cooking-support modules are specified so that what the system sensed, what it proposes, and how the user can revise it appear on the shared surface, following human-AI interaction guidance on status, correction, and control.
- **A two-study empirical arc.** User Study 1 grounds the requirements; User Study 2 (N = 36) shows the developed direction preferred on every premium-experience attribute, with the margin confirmed by

a respondent-level sign test, while locating the remaining work in first-use onboarding and routine simplicity.

2. Related Work

2.1. Premium experience in use

Experience-design research distinguishes product qualities from the experience that emerges through use. [Hassenzahl \(2010\)](#) argues that experience arises from situated interpretation and meaningful action, not from functionality alone, and that hedonic value rests on pragmatic foundations. For a premium oven, finish and feature count create expectations that must be sustained during everyday cooking. A calm exterior therefore needs to be matched by an equally clear experience in use.

2.2. Legibility, feedback, and control

[Norman \(2013\)](#) treats visibility, feedback, mapping, and signifiers as the conditions for a workable conceptual model. Cooking stresses all four at once, because the task is embodied and time-bound: heat, steam, tray geometry, and doneness are only partly observable, and mistakes are costly in food and time. [Kim et al. \(2023\)](#) pursue a closely related goal for Korean lightwave ovens, redesigning the interface around a touchscreen to improve the cooking experience. Our project keeps that focus and moves it onto the product itself: the front surface is treated as the place where digital state and physical action must meet.

2.3. Human control in domestic AI

Guidelines for human-AI interaction ask systems to make capabilities clear, show status, support efficient correction, and recover gracefully ([Amershi et al., 2019](#)). [Shneiderman \(2020\)](#) argues that high automation and high human control should be designed together. These principles are particularly relevant in a domestic kitchen. People may welcome a recommendation, but they still season to taste, intervene during cooking, and judge results against household preferences. The Panorama-Display Lightwave Oven therefore keeps AI output visible at the point of use: each module presents a proposal on the Panorama Display that the cook can accept, edit, ignore, or rate.

3. Methods

3.1. Research process

The project followed the Double Diamond model: a first diamond from discovery to problem definition, and a second from concept development to evaluation ([Design Council, 2019](#)). Figure 2 shows how the two user studies sit in that frame. User Study 1 produced the diagnosis and the design requirements; User Study 2 tested whether the developed direction answered them. The two studies were planned as one arc, so the evaluation criteria in the survey are the same qualities the formative work found missing.

3.2. User Study 1: formative study

User Study 1 was conducted in the midterm phase with six participants recruited along a cooking-experience spectrum, following the logic of small-sample usability work ([Nielsen and Landauer, 1993](#); [Nielsen, 2000](#)). All six completed in-depth interviews. Three first-time users additionally completed cooking task scenarios with concurrent think-aloud ([Ericsson and Simon, 1993](#)); first-time users were chosen for the task track because habitual users absorb a product’s quirks into routine and stop showing the friction the tasks were meant to surface. Sessions ran from April 4 to April 13, 2026, lasted about 90 minutes, and were audio- and video-recorded with verbal consent; participants could withdraw at any time. Table 1 summarizes the group.

The study ran as three streams, each answering a different question. **Stream 1A: external interviews and co-design.** Experienced users (V1–V3) discussed purchase background, real cooking patterns, parameter decisions, automatic cooking, attempts at new recipes, ThinQ use, unexpected outcomes, and what an ideal oven would look like. Each session closed with mind mapping, brainstorming, idea selection, and rapid paper prototyping ([Lucero, 2015](#)), so participants could externalize preferred interface structures instead

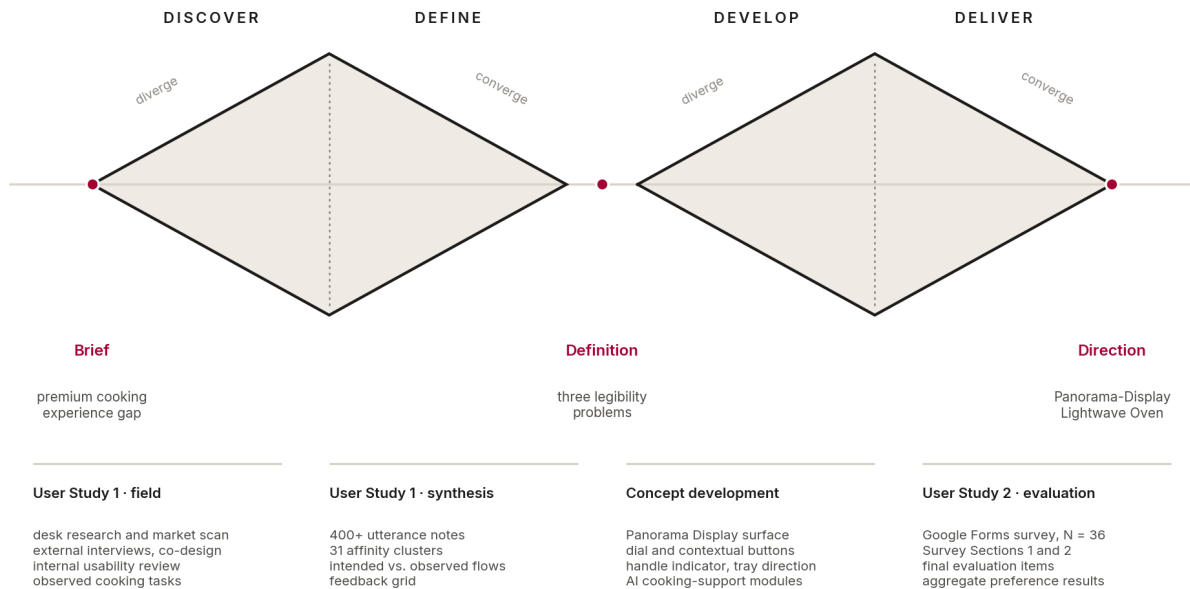


Figure 2. Design research process. The first diamond moved from field discovery to three legibility problems; the second developed the Panorama-first concept and evaluated it against a comparison design.

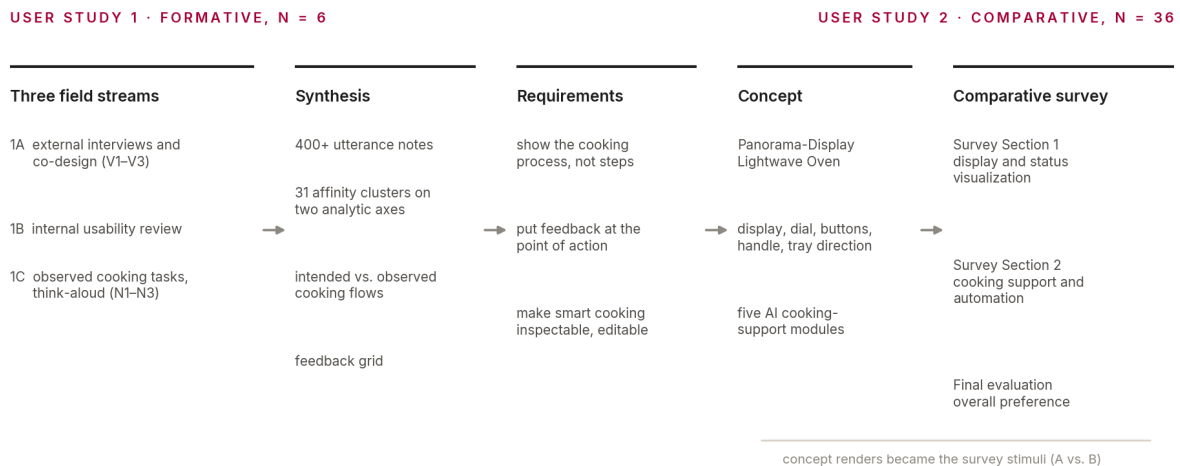


Figure 3. Integration of the two studies. User Study 1 streams converge into requirements; the concept built on those requirements supplies the stimuli and criteria for User Study 2.

of only naming complaints. **Stream 1B: internal usability review.** The research team's internal protocol walked the existing oven's entry points, dial-and-button sequence, display constraints, tray actions, and maintenance steps, recording where the official structure diverged from a newcomer's expectations. **Stream 1C: observed cooking tasks.** The three first-time users cooked while thinking aloud: N1 and N2 prepared oven-baked chicken and then frozen dumplings through automatic cooking; N3 prepared steamed pork belly using the supplied manual and a separate search device. We recorded mode discovery, time and temperature interpretation, manual consultation, hesitation, backtracking, emotional state, and the points where each participant intervened or recovered. The full protocols, including the worksheet materials, are preserved in the Supplementary Material (Appendix A).

Table 1. User Study 1 participants. V denotes experienced lightwave-oven users in the interview track; N denotes first-time users in the usability-task track.

ID	Track	Profile	Cooking and oven experience
V1	Interview	Female, 50s, homemaker	About five years of LG Lightwave use; twenty years of homemaking
V2	Interview	Female, 30s, PR/marketing	Two to three years of LG Lightwave use; amateur baking
V3	Interview	Male, teens, dormitory	One to two years of LG Lightwave use; self-cooking
N1	Usability task	Male, 20s, living alone	Microwave-only cook with limited cooking experience
N2	Usability task	Male, 20s, dormitory	Beginner home cook; bartending background
N3	Usability task	Male, 30s, living alone	Enthusiastic home cook; other-oven experience, new to this model

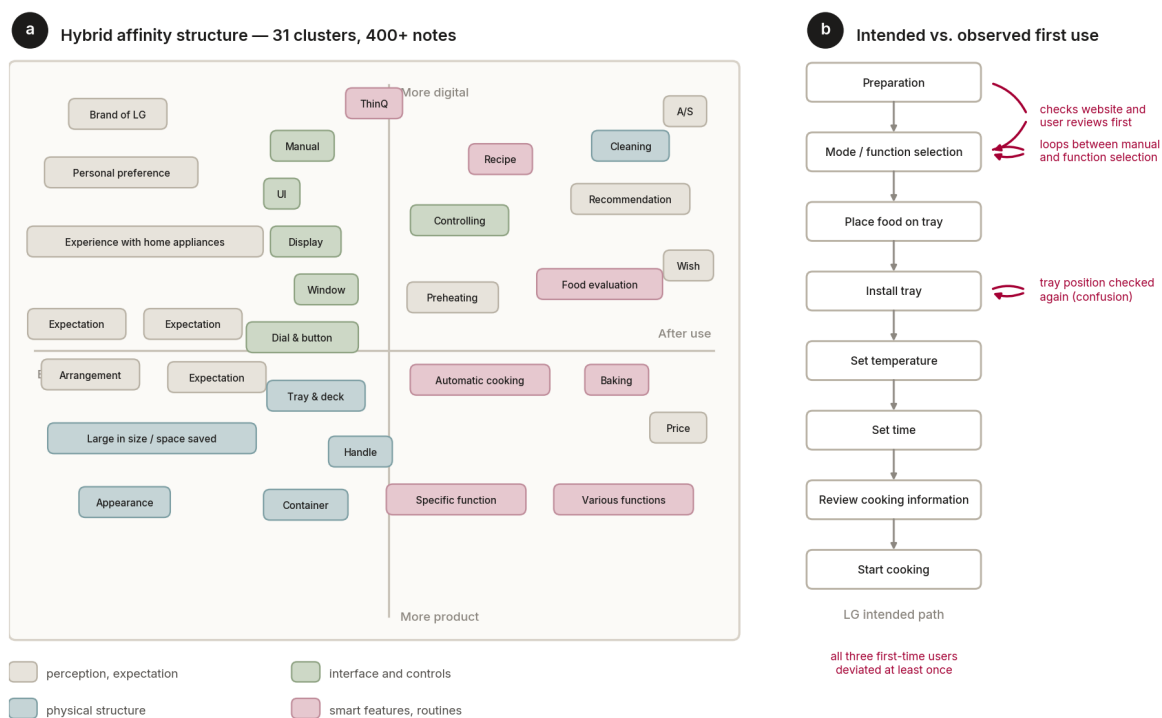


Figure 4. Formative evidence, redrawn from the originals in Appendix C. (a) The hybrid affinity structure: 31 clusters from 400+ notes on two analytic axes. (b) Structural summary of the intended-versus-observed flow comparison; every first-time user left the intended path at least once, most often around mode selection and tray setup.

3.3. Analysis of the formative data

Transcripts and field notes were turned into more than 400 sentence-level sticky notes and clustered by five researchers into a hybrid affinity diagram, extending the contextual-design tradition (Beyer and Holtzblatt, 1998). We call the diagram hybrid because two analytic axes were fixed deductively, a horizontal Before-Use to After-Use axis for where in the cooking journey an issue appeared and a vertical axis from digitally mediated to physically mediated concerns, while the 31 keyword clusters formed inductively on that canvas. Disagreements over placement were resolved in discussion. Two further tools supported interpretation: a flow analysis comparing the operation sequence in the official LG tutorial (LG Electronics Service, 2024) with the sequences N1–N3 actually followed, and a feedback grid sorting the co-design output into what worked, what needed improvement, open questions, and ideas. Figure 4 presents both analyses redrawn for print legibility; the original affinity board and the complete per-task flow charts appear in the Supplementary Material (Appendix C).

3.4. Concept development

The synthesis produced three design requirements, stated here in the order the concept addresses them: show the cooking process as a visible sequence rather than isolated confirmations; place feedback at the point of physical action, especially around trays, steam, and the open door; and make smart cooking inspectable and editable instead of hidden behind presets. The team developed the Panorama-Display Lightwave Oven against these requirements through sketching, interface specification, and rendering, and prepared a compact-display comparison design that keeps a conventional small screen and familiar physical structure. The comparison design exists for evaluation: it represents the path the product line would take without the Panorama-first commitment.

3.5. User Study 2: comparative concept survey

User Study 2 evaluated the developed direction against the comparison design through a Google Forms survey with 36 respondents, collected between June 2 and June 8, 2026. The survey used a within-subject structure: every respondent saw both directions, labeled Concept A (the developed direction) and Concept B (the comparison design) in the form. The product name was not revealed during the survey, so respondents compared the stimuli rather than a brand story; in this report the developed direction is presented under its design name, the Panorama-Display Lightwave Oven, the labels are abbreviated to A and B, and the label A is used only when describing the survey condition. The sample skewed young: of the 35 respondents with recorded background, 27 were in their twenties, three in their thirties, one in their forties, three fifty or older, and one under twenty. Estimated completion time was five to seven minutes. Participation was anonymous except for an optional phone-number field for a raffle; among respondents who opted in, twenty were randomly selected to receive a 5,000 KRW Naver Pay voucher.

The form opened with study information, a consent item, and six background items covering age group, living arrangement, cooking frequency, experience with oven-type appliances, tendency to try new recipes, and interest in smart-appliance features. Two stimulus sections followed. **Survey Section 1** compared display and cooking-status visualization using a front-view image pair and an open-door image pair (Figure 5); its six forced-choice items covered current-status understanding, temperature and time checking, clarity of the next action, ease of first-time learning, kitchen convenience, and premium feel. **Survey Section 2** compared cooking support and automation using a cooking-support image pair; its six forced-choice items covered pre-cook uncertainty, setting burden, confidence for new recipes, mistake reduction, expected regular use, and overall smartness, followed by a multiple-selection item on which elements of A seemed useful and an open-ended reason. A **final evaluation** then asked respondents, with both sections in view, to choose between the directions on overall experience, process understanding, failure anxiety, willingness to try new recipes, and real purchase consideration, to rate purchase intent for each direction on an eleven-point scale, and to make a final forced choice with a written reason. Every forced-choice item offered three options: A, B, and Similar / Not sure. The complete instrument, including item wording and response formats, is reproduced in the Supplementary Material (Appendix B). Section 6 reports the results computed from the anonymized response-level export, with the extended analysis in Appendix D. One response reached the export without its Section 1 grid and background fields, so Section-1-based summaries use $n = 35$ and all other summaries use $n = 36$.



Figure 5. Visual stimuli used in User Study 2, shown to respondents as A and B. Survey Section 1 used the front-view and open-door pairs; Survey Section 2 used the cooking-support pair. The original combined images from the form are preserved in Appendix B.

4. Results: User Study 1

The formative data describe one connected experience gap rather than three separate flaws. Figure 6 maps the chain from problem to requirement to concept response.

Problem 1: minimalist control, layered hierarchy. The dial and buttons are individually simple, and their visual restraint is a deliberate part of the Objet identity. The cost appears underneath: mode, function, weight, time, and temperature are compressed onto one dial and a small display, so the user passes through several selection layers with little confirmation of where they are. Participants paused mid-task to ask whether a dial press had registered, struggled to separate oven, microwave, and air-fryer modes without the manual, and often restarted after a wrong turn. All three first-time users left the tutorial's intended sequence at least once, most often at function selection and dial confirmation. The minimalist exterior therefore shifted complexity into memory and repeated confirmation during use.

Problem 2: physical setup weakens trust. Tray logic, heat distribution, steam preparation, and cleaning shaped judgments as strongly as the interface did. V1 and V2 described heating as uneven and tray choice as guesswork despite the manual's assignments. N3's steam task made the misfit concrete: detaching, filling, and reattaching the water tank, repositioning the tray, and returning to the interface sequence were attempted out of order because the physical steps and the digital steps gave no cues to each other. Maintenance (odors, ridged trays, the reservoir) felt heavier than a premium price implies.

Problem 3: premium functions do not become habits. Automatic cooking, recipe scanning, and ThinQ were the features most tied to the product’s positioning, and the least absorbed into routine. Participants defaulted to manual mode, fetched recipes elsewhere, and could not say what ThinQ added; V1 asked directly why it existed. The resistance was not to automation as such. People could not see what a program was about to do, how it was structured, or how much room they had to intervene once it began, and under that opacity manual control felt safer. Several of the most pointed design ideas, among them progress made visible, voice support, internal-state feedback, and personalized recipes, surfaced only in the prototyping phase, after the verbal interview had closed.

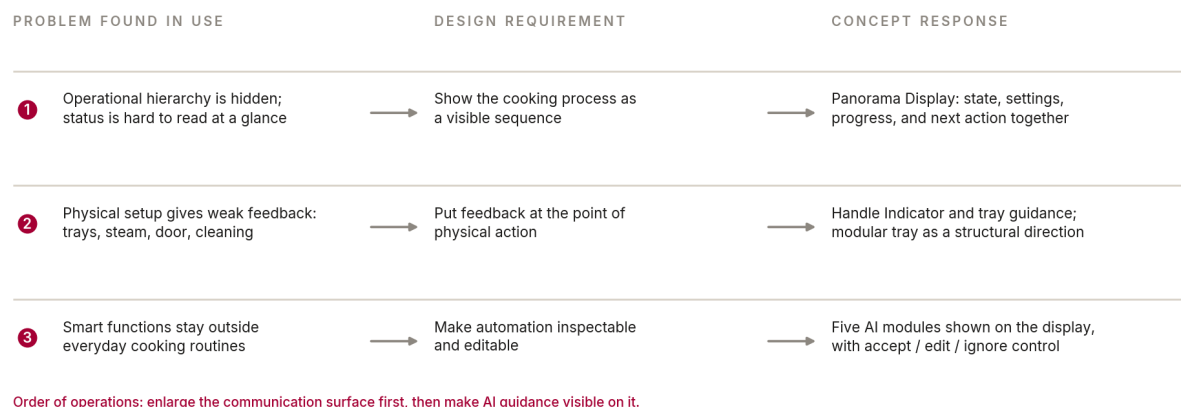


Figure 6. From problems to requirements to concept responses. The order matters: the communication surface is enlarged first, and AI guidance is then made visible on it.

5. Results: Design

The final concept is organized as a product-interface system rather than a list of separate features. The Panorama Display provides the shared surface; the dial, digital buttons, handle indicator, and tray direction keep action direct; and the AI layer uses the same surface for recommendations and feedback. Each part responds to a specific formative finding.

5.1. Panorama Display

The Panorama Display extends the information surface across the full oven window as a transparent 480×55 band, so the cook reads the process over the food instead of beside it. This is the primary design decision, and it responds to Problem 1: a conventional small display can confirm one value at a time, while a cooking session needs mode, recipe, remaining time, sensed weight, target temperature, stage position, warnings, and the next action available together. The display is organized in three layers (Figure 7). The *state layer* keeps mode, recipe, time, weight, and temperature persistently visible. The *process layer* runs a horizontal stage timeline (preparation, first cook, mid-check, second cook, finish), so the user always has a position in the sequence. The *recommendation layer* attaches AI proposals to the parameters they concern, labeled as proposals and editable in place. Because the band is transparent, vision-based cues can sit directly over the visible food.

5.2. Direct control and handle-area feedback

The formative work did not identify the dial itself as the main problem; the difficulty came from the hidden hierarchy around it. The concept therefore keeps a single physical dial as the stable tactile anchor: turn to browse, press to confirm, press left or right to move between stages now that the whole sequence is visible. The fixed button bank is removed. Contextual digital buttons appear only when a choice is live:

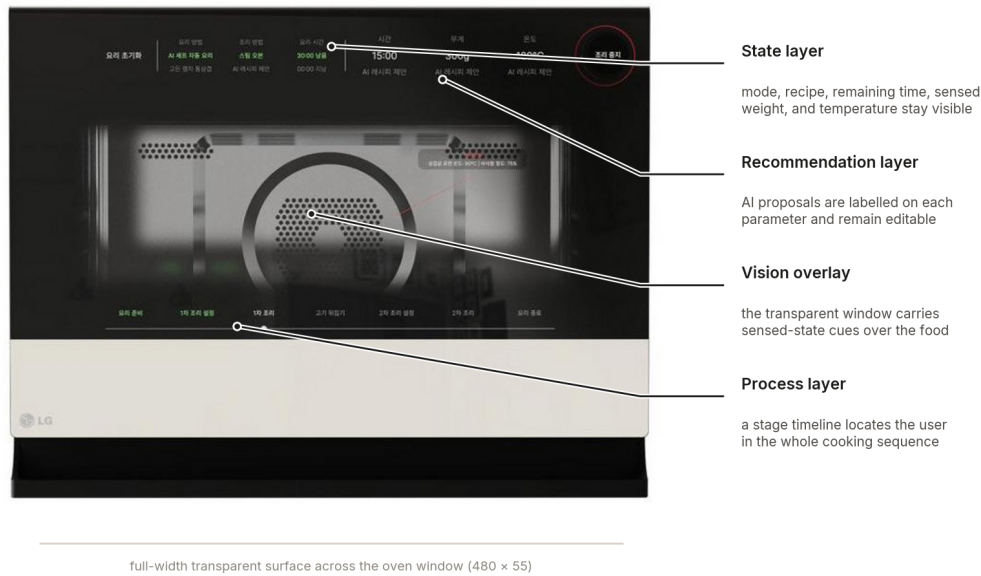


Figure 7. Panorama Display detail on the developed front view. State, recommendation, vision, and process layers share one surface; AI proposals are labeled on the parameters they would change.



Figure 8. Handle-area feedback and the control model. The handle indicator continues guidance during open-door actions; the single dial with contextual digital buttons replaces the fixed button bank.

selecting a method, accepting a proposed setting, skipping a step, stopping a cook. This preserves direct manipulation while reducing persistent button clutter.

The handle indicator answers Problem 2 at its sharpest point. Many risky moments happen while the main display is out of view: steam release, tray insertion and removal, turning food, judging whether surfaces are safe to touch. A slim display along the door's top edge keeps warnings, tray cues, and the immediate next action at the point of contact while the door is open (Figure 8). The same problem group motivates the modular tray direction: trays whose configuration matches the selected mode, reducing the guesswork V1 and V2 described. In this report the tray remains a structural direction rather than a finished mechanism, and we mark it as such.

5.3. The AI layer on the display

The AI layer addresses Problem 3 by keeping system activity visible. Five cooking-support modules are specified, and every one of them renders its work on the Panorama Display: what was sensed, what is proposed, what will change, and where the user can step in (Figure 9, Table 2). Voice Chef takes hands-free commands when fingers are floured or oily. Auto Cook Agent reads ingredient state and weight from the internal camera and sensors and proposes mode, temperature, time, and tray placement. Chef Persona converts a chef’s or creator’s recipe into staged settings and situated coaching. Vision Feedback reads doneness and surface state and prompts turning, checking, or early exit. Taste Memory stores post-cook preference, from crispness and doneness to per-family-member profiles, and adjusts the next session. All five modules follow the same user-control loop: each output is a proposal that the cook can accept, edit, ignore, or rate by dial, touch, or voice. The wider surface keeps these automated decisions open to review and adjustment instead of reducing them to a beep or hidden preset.

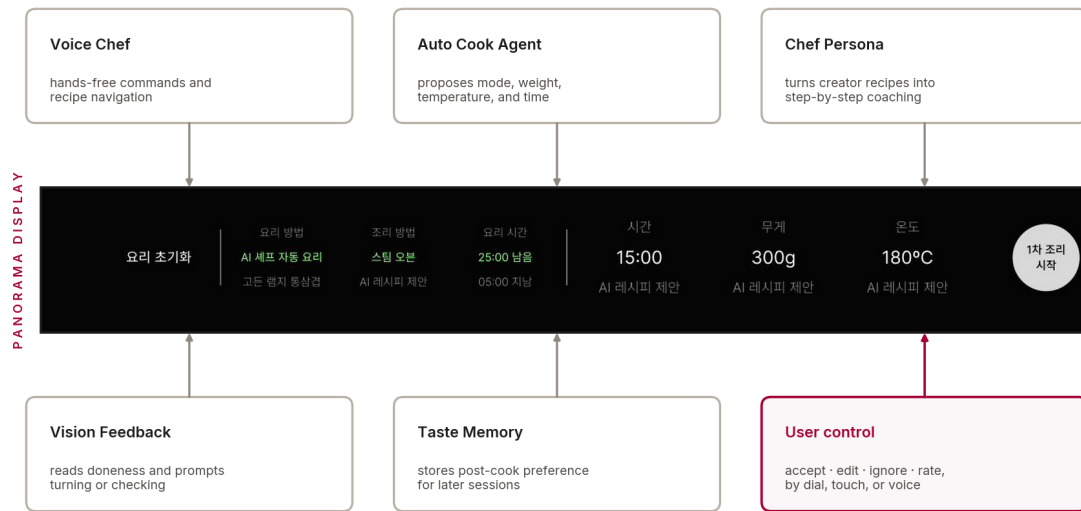


Figure 9. The AI system map. Five cooking-support modules and the user-control loop meet on the Panorama Display, shown here with the actual state band from the working renders.

Table 2. Cooking-support modules in the Panorama-Display Lightwave Oven.

Module	Role	Information shown on the Panorama Display	Available user control
Voice Chef	Hands-free command and recipe navigation	Spoken intent translated into mode, time, temperature, and the next step	Confirm, cancel, repeat, or revise by voice or dial
Auto Cook Agent	Automatic setting support from camera and weight sensing	Proposed mode, temperature, time, weight, and tray placement, with each item labeled as a proposal	Apply, fine-tune, or switch to manual
Chef Persona	Creator-style recipe coaching	Recipe source, staged plan, and the reason for each step	Choose a coaching style or reduce guidance
Vision Feedback	Camera-based state reading during cooking	Surface-state readout, turning prompts, and exit suggestions	Accept the cue, check manually, or delay
Taste Memory	Preference learning across sessions	Post-cook rating, saved household profiles, and next-session adjustment	Rate, edit, or reset the stored preference

5.4. Illustrated cooking scenario

Figure 10 walks one session through the working interface renders. The user picks a cooking method or hands the choice to the AI chef; Chef Persona presents a recipe as a staged plan with settings attached; the proposed time, weight, and temperature are confirmed or tuned with the dial; during the cook the timeline advances and Vision Feedback posts the sensed surface state; at the flip point the prompt names the action and the moment; near the end the system offers an early exit for a softer finish; after the cook a one-turn rating feeds Taste Memory. Nothing in the sequence requires the user to trust a hidden program. Every machine decision arrives as something on screen that can be changed.

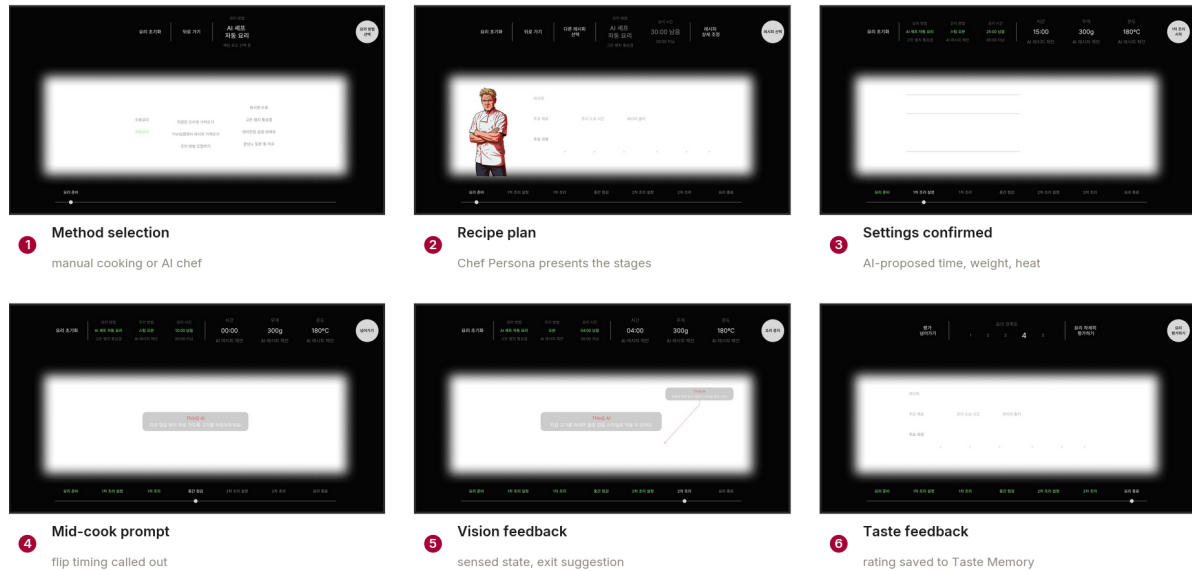


Figure 10. One cooking session on the Panorama Display, from the working renders. The same surface carries method choice, the staged recipe plan, confirmed settings, the mid-cook prompt, vision feedback with an exit suggestion, and the post-cook rating.

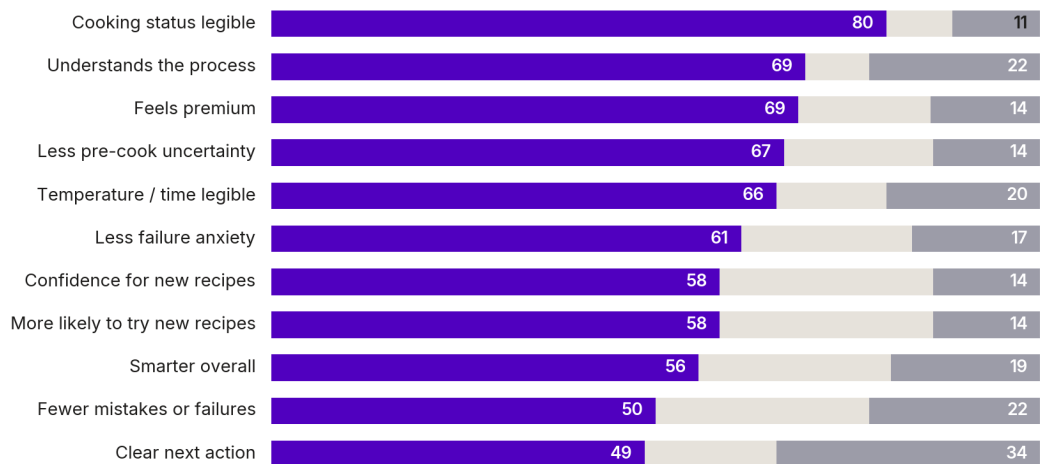
6. Results: User Study 2

6.1. The developed direction led across the premium-experience attributes

Across the eleven attributes tied to the project's premium-experience goal, the developed direction was the most-chosen option on every one (Table 3, Figure 11). The strongest margins appeared in the areas identified by the formative study: cooking-status legibility (80% vs. 11%), process understanding (69% vs. 22%), premium feel (69% vs. 14%), reduced pre-cook uncertainty (67% vs. 14%), and temperature and time legibility (66% vs. 20%). The confidence-related items showed the same pattern: reduced failure anxiety (61% vs. 17%), confidence for new recipes and willingness to try them (58% vs. 14% each), a smarter overall experience (56% vs. 19%), and fewer expected mistakes (50% vs. 22%). The section-level preference followed the same direction: asked for an overall preference within Survey Section 1, 57% chose the developed direction against 40% for the comparison design.

Summarized as a single index, respondents selected the developed direction on a mean of 6.7 of the eleven premium attributes, against 2.1 for the comparison design and 2.2 marked Similar / Not sure (Figure 12). The respondent-level pattern was consistent with the aggregate result: 29 of the 35 respondents with complete grids chose the developed direction on more premium attributes than the comparison design, five showed the reverse pattern, and one tied. An exact sign test confirmed this difference ($p < .001$), indicating that the preference was not produced only by pooled percentages.

PREMIUM-EXPERIENCE ATTRIBUTES



FIRST-USE AND ROUTINE ATTRIBUTES

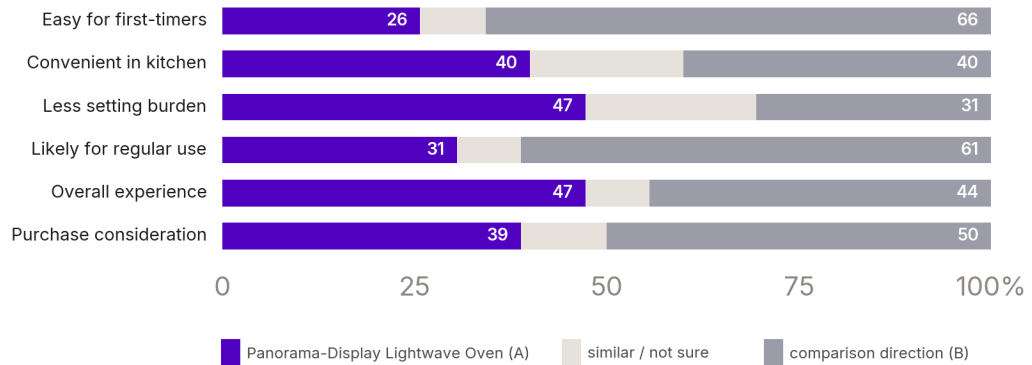


Figure 11. Forced-choice shares by attribute (N = 36; Survey Section 1 rows use n = 35, see Table 3). Premium-experience attributes are sorted by the developed direction's share; first-use and routine attributes keep the instrument's order.

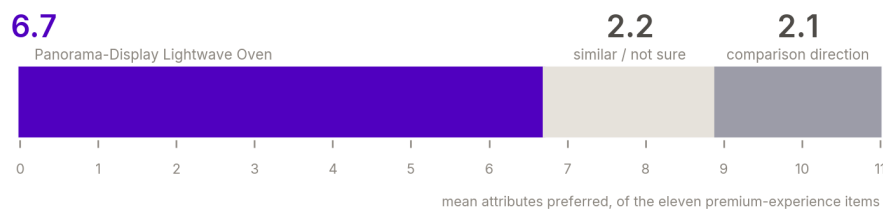


Figure 12. Premium-experience index: mean number of the eleven premium attributes on which each option was selected, over the 35 complete response grids. 29 of 35 respondents favored the developed direction on more attributes than the comparison design (exact sign test, $p < .001$).

6.2. The comparison design remained stronger for familiar routines

The comparison design led where familiarity mattered most: ease for first-time users (66% vs. 26%) and expected regular use (61% vs. 31%). The other routine-use items were more mixed. Kitchen convenience was evenly split (40% each), while the developed direction led on reduced setting burden (47% vs. 31%) and held a small advantage on overall experience (47% vs. 44%). The comparison design retained a lead on the categorical purchase question (50% vs. 39%), but the eleven-point purchase-intent ratings were much closer: means of 5.3 for the developed direction and 5.5 for the comparison design, medians of 6 and 5, with 16 respondents rating the developed direction higher, 15 rating the comparison design higher,

Table 3. Forced-choice results from User Study 2. Values are counts with rounded row percentages in parentheses. Survey Section 1 rows use n = 35 because one response did not contain that grid; all other rows use n = 36.

Attribute	Developed (A)	Similar / not sure	Comparison (B)	n
<i>Premium-experience attributes</i>				
Cooking status legible	28 (80%)	3 (9%)	4 (11%)	35
Understands the process	25 (69%)	3 (8%)	8 (22%)	36
Feels premium	24 (69%)	6 (17%)	5 (14%)	35
Less pre-cook uncertainty	24 (67%)	7 (19%)	5 (14%)	36
Temperature / time legible	23 (66%)	5 (14%)	7 (20%)	35
Less failure anxiety	22 (61%)	8 (22%)	6 (17%)	36
Confidence for new recipes	21 (58%)	10 (28%)	5 (14%)	36
More likely to try new recipes	21 (58%)	10 (28%)	5 (14%)	36
Smarter overall	20 (56%)	9 (25%)	7 (19%)	36
Fewer mistakes or failures	18 (50%)	10 (28%)	8 (22%)	36
Clear next action	17 (49%)	6 (17%)	12 (34%)	35
<i>First-use, routine, and purchase attributes</i>				
Easy for first-timers	9 (26%)	3 (9%)	23 (66%)	35
Convenient in kitchen	14 (40%)	7 (20%)	14 (40%)	35
Less setting burden	17 (47%)	8 (22%)	11 (31%)	36
Likely for regular use	11 (31%)	3 (8%)	22 (61%)	36
Overall experience	17 (47%)	3 (8%)	16 (44%)	36
Purchase consideration	14 (39%)	4 (11%)	18 (50%)	36

and five tied. Taken together, these ratings suggest parity in anticipated willingness to pay; the categorical item may reflect the perceived safety of a familiar choice rather than a rejection of the developed concept. One section-level result differed from the item-level pattern. When asked for an overall preference within Survey Section 2, 53% chose the comparison design and 39% chose the developed direction, even though the section's individual items on uncertainty, confidence, mistakes, and smartness favored the developed concept. The open-ended responses help explain this difference: respondents valued what the support could do but were cautious about how much information appeared at once. This points to information volume and timing as design refinements rather than to a rejection of visible cooking support.

6.3. Open-ended responses

The open-ended responses help explain the numerical pattern. The most frequent reason for choosing the developed direction was the ability to see both the food and the cooking process. More than ten respondents grounded their choice in visibility—"the inside is clearly visible, so it is easy to check how the food is coming along," "you can confirm the cooking state with your own eyes," "a design that lets you take in the whole cooking process at a glance." A second cluster read the surface as status: the developed direction felt "newest," "trendy," premium enough to justify a higher price. A third, smaller cluster named the support itself: voice guidance for cooks who do not know how to cook, the sense of participating in the process rather than waiting for it, and knowing when to step in.

Comments favoring the comparison design concentrated on three themes. Several respondents preferred physical buttons because of durability and one-button-one-function operation; several warned about information volume ("only the information I need," "too much input is confusing"); and a few anticipated fatigue from an assertive or repetitive AI voice. One respondent summarized the routine-use concern directly: most sessions involve familiar food, so the easier-to-use machine wins. These concerns focused on quantity, tone, and habit, and therefore point to how the concept should adapt rather than to a rejection of visible feedback.

6.4. Final concept synthesis

Taken together, the study findings reinforce the final concept as two coordinated layers rather than a collection of independent features. The product-interface layer makes the cooking process visible and keeps control at the point of action through the Panorama Display, the physical dial with contextual digital buttons, and the handle indicator. The AI cooking-support layer uses the same surface for voice guidance, staged recipe coaching, and vision-based feedback. Figure 13 summarizes the six touchpoints shown in dedicated render states. Auto Cook Agent and Taste Memory remain part of the wider system described in Table 2; in this overview, they are expressed through the shared display rather than as separate panels.

PRODUCT INTERFACE

Shared surface, direct control, and feedback at the point of action.



01 Panorama Display

Displays settings, progress, and guidance across the oven glass.



02 Single Dial + Contextual Buttons

Keeps tactile control precise while digital buttons appear only when relevant.



03 Handle Indicator

Provides next-step guidance right where you hold the door.

AI COOKING SUPPORT

Visible sensing and coaching that remain inspectable and editable.



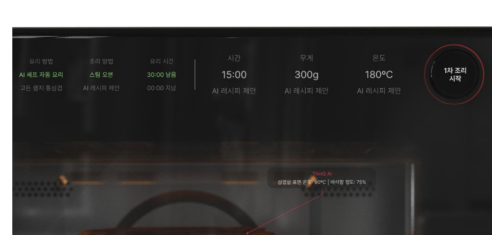
01 Voice Chef

Translates spoken intent into settings, navigation, and immediate next steps.



02 Chef Persona

Turns creator recipes into staged coaching with editable parameters.



03 Vision Feedback

Shows sensed doneness, turning points, and exit suggestions in real time.

Figure 13. Final concept synthesis. Product-interface features (left) and AI cooking-support features (right) are coordinated through the shared Panorama Display. Auto Cook Agent and Taste Memory are specified in Table 2 and operate through the same display states.

7. Discussion

7.1. The display enables credible domestic AI

The project began with cooking-support AI as an attractive direction, and the studies clarified an important order of operations: the interaction surface comes first. Cooking is visible, physical, and interruptible. The formative findings showed that participants hesitated when automation could not be inspected, rather than rejecting automation itself. On the Panorama Display, the five modules are presented as visible proposals, staged progress, and sensed evidence instead of isolated beeps or hidden presets. Survey Section 2 supports this direction: respondents associated the developed concept with less pre-cook uncertainty, fewer expected mistakes, and a smarter overall experience. These findings suggest that the display and AI support should be designed as one system. The display makes automated guidance understandable and editable, while the AI layer gives the expanded surface a clear role throughout cooking.

7.2. Premium cooking depends on confidence

The developed direction's largest margins—status legibility, process understanding, reduced uncertainty, and reduced failure anxiety—all relate to confidence. This aligns with the formative finding that participants judged the oven by how safely they could act, not simply by how many functions it offered. In this concept, a premium appliance supports confidence before cooking by showing what will happen, during cooking by showing where the process stands, and after cooking by showing what can be adjusted next time. Exterior quality establishes a premium expectation; legibility sustains that expectation during use.

7.3. Balancing rich guidance with everyday calm

The routine-use results and open-ended responses point to a clear refinement rather than a change of direction. Respondents valued visibility but were cautious about information volume and persistent guidance. The next iteration should therefore make the Panorama Display adaptive: a calm default that shows only essential information, expansion when uncertainty increases or the user asks for more detail, and short direct paths for reheating and other simple tasks. This approach preserves the premium-experience advantage demonstrated in the survey while addressing the first-use and everyday-use concerns captured by the comparison design. The dial's stage-stepping behavior already supports this balance because the visible sequence reduces the need for separate back navigation.

8. Limitations and Future Work

The evaluation used visual concept stimuli, not a working oven. The results therefore measure anticipated experience; they cannot show whether users accept, revise, or ignore AI proposals under heat, mess, and time pressure, or how the display reads across a kitchen at a glance. An interactive prototype in real cooking sessions is the necessary next step, and it should test the adaptive-calmness behavior proposed above.

The AI modules are design specifications. Ingredient recognition, weight estimation, doneness reading, recipe-to-setting translation, and preference learning each require technical validation before any implementation claim, and the modular tray remains a structural direction without a mechanism. The survey sample ($N = 36$) suits a design-methods course evaluation, not market inference, and it skews young: 27 of the 35 respondents with recorded background are in their twenties, so the views of older cooks—two of whom, notably, chose the developed direction citing the readability of a small display—are thinly represented. One response reached the export without its Section 1 grid, so Section-1-based summaries rest on 35 responses; the analysis otherwise uses the full anonymized export (Appendix D). The statistics reported here are deliberately simple—counts, shares, and exact sign tests—because the design questions at stake are directional; the multiple-selection item on useful elements was collected but is not analyzed in this report. Future evaluation should keep row-level data with consent, code the open-ended reasons, and add repeated-use measures so that the link between legibility, trust, routine adoption, and purchase consideration can be tested rather than described.

9. Conclusion

This report reframed a premium cooking appliance around one question: can the cook read the process? User Study 1 located the existing oven's gap in three legibility problems—hidden operation hierarchy, weak feedback during physical actions, and smart functions that did not become part of routine use. The Panorama-Display Lightwave Oven responds in the same order: a full-width transparent display that carries state, progress, and proposals over the food; a dial, contextual buttons, handle indicator, and tray direction that keep action direct; and five AI cooking-support modules whose work remains visible and correctable on the shared surface. User Study 2 showed that the direction was preferred on every premium-experience attribute it targeted, while also identifying the need to keep the richer surface calm enough for everyday use. In this concept, premium experience depends not only on a minimal exterior but also on making the cooking process visible and understandable.

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Supplementary Material

The following appendices preserve the study instruments and the original project visuals so that the work in the main text can be traced and repeated. Appendix A reproduces the User Study 1 protocols, Appendix B the complete User Study 2 survey instrument and stimuli, Appendix C the original working visualizations, and Appendix D the aggregate evaluation data included in the report package.

A. Survey Protocols: User Study 1

User Study 1 ran as three separate streams. The external interview and co-design protocol and the internal usability and observed cooking-task protocol are reproduced here in full; Stream 1B used the same session frame as the observed tasks with the research team as operators.

A.1. External interview and co-design protocol

Table 4. External interview and co-design procedure (experienced users V1–V3).

Stage	Time	Procedure
Introduction	1–2 min	Explain the purpose, anonymity, research use of responses, voluntary participation, and audio recording. Confirm consent before continuing.
Interview	15–20 min	Ask about purchase background and expectation; actual use patterns and parameter decisions; automatic-cooking experience; new-recipe attempts; ThinQ app use; unexpected outcomes; and overall evaluation.
Mind map	5 min	Ask the participant to write words, emotions, situations, good points, and pain points around the prompt “LG lightwave oven user experience.” Photograph the sheet and transcribe repeated keywords.
Brainstorming	5–7 min	Turn mind-map issues into solution ideas. Probe desired information, reassuring feedback, automatic-function trust, interface structure, and control preference.
Idea selection	2–3 min	Ask the participant to choose one or two ideas that would resolve the largest inconvenience.
Rapid prototype	12–15 min	Using the oven-front template, screen template, button and dial pieces, Post-its, scissors, glue, and pens, build a low-fidelity interface or control layout.
Wrap-up	2–3 min	Ask for final comments and anything not yet covered.

The interview modules were ordered as follows: purchase background and expectation; actual use pattern; automatic-cooking experience; new-recipe attempts; ThinQ app experience; unexpected situations; and overall evaluation. Representative questions asked why the product was purchased, which cooking modes were used most often, how time and temperature were decided, whether automatic cooking was trusted, why new recipes were or were not attempted, which ThinQ features were used, how unexpected results were handled, and whether the participant would recommend the product.

A.2. Internal usability and observed cooking-task protocol

Table 5. Internal usability and observed cooking-task procedure (first-time users N1–N3).

Stage	Time	Procedure
Introduction and consent	3 min	Explain that the participant is not being tested; the goal is to understand the product experience. Confirm video-recording consent.
Pre-interview	5 min	Ask about appliance experience, cooking frequency, brand impression, oven familiarity, and expectations for the product.
Context setting	1 min	Place the participant in a realistic dinner-preparation situation. Allow manual use if needed, but ask for continuous think-aloud.
Task 1: oven-baked chicken	15–20 min	Ask the participant to begin cooking the prepared chicken. Observe mode discovery, time and temperature interpretation, manual consultation, hesitation, backtracking, emotion, and start success.
Interview 1	5 min	Ask about first-use impression, difficulty, perceived premium value, willingness to use again, and willingness to pay for improvement.
Task 2: frozen dumplings	15–20 min	Ask the participant to use automatic cooking for frozen dumplings. Observe automatic-menu discovery, trust, package checking, tray checking, and perceived control.
Interview 2	10 min	Ask about automatic-function trust, overall satisfaction, failure anxiety, and improvement needs.
Co-design sequence	15–20 min	Conduct mind mapping, brainstorming, idea selection, and rapid prototyping with the same logic as the external protocol.
Wrap-up	3–5 min	Ask for final feedback about the session and the most important change needed.

N3's session replaced the two tasks above with a steamed pork belly task using the supplied manual and a separate search device, which exposed the water-tank and tray sequence reported in Section 4. Planned data across the stream included task success, task time, error count, help requests, manual references, backtracking actions, satisfaction and difficulty ratings, automatic-function trust, reuse intention, think-aloud records, interview responses, confusion-point notes, mind maps, brainstorming ideas, and rapid prototypes.

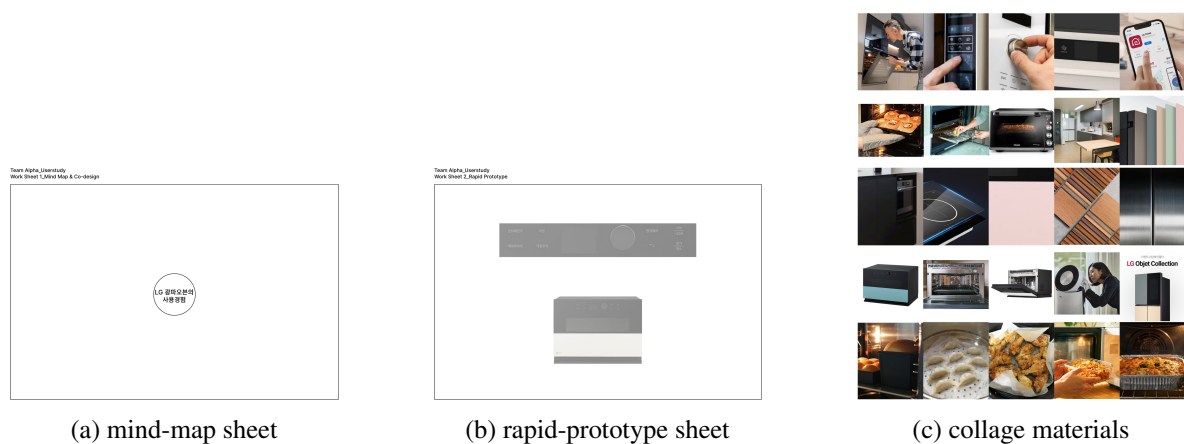
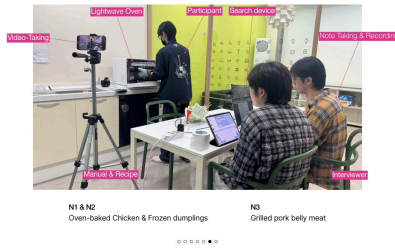


Figure 14. User Study 1 worksheet materials, reproduced upright and separated: (a) the mind-map sheet, (b) the rapid-prototype sheet with the oven and control-panel templates, and (c) the magazine and photographic pieces provided for collage.

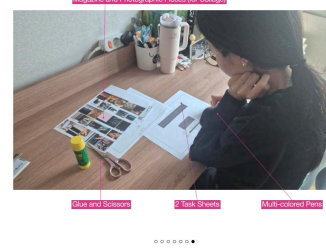
User Study | Task Scenario + Think Aloud

A task scenario is a specific context and goal provided to a user, while think-aloud is a protocol where users verbalize their thoughts in real-time as they perform that task.



User Study | Mind Map & Rapid Prototype

A mind map is a visual diagram used to hierarchically organize information around a central concept, while rapid prototyping is a fast-paced technique used to quickly create tangible models for early testing and validation of ideas.



(a) observed cooking task with think-aloud

(b) mind-mapping and rapid-prototyping session

Figure 15. Session documentation from the original midterm materials, reproduced as archived.

B. Survey Protocol: User Study 2

The comparative concept survey was administered through Google Forms as the *LG Lightwave Oven Concept Comparison Survey*. The form introduced the activity as a comparison between two lightwave-oven interaction directions, labeled Concept A and Concept B, and did not reveal their design background; this report abbreviates the labels to A and B. Respondents were asked to inspect the images and imagine using each direction in a real cooking situation. Estimated completion time was 5–7 minutes. Responses were anonymous except for an optional phone-number field for the raffle; twenty randomly selected opt-in respondents received a 5,000 KRW Naver Pay voucher.

B.1. Information, consent, and background

The first page stated the study purpose, the approximate completion time, anonymity, the optional raffle, and voluntary participation, followed by a consent item. Six background items followed (Table 6).

Table 6. Participant-background items.

Question	Response format
Age group	Under 20; 20s; 30s; 40s; 50 or older.
Current living arrangement	Living alone; living with spouse or partner; living with family; other.
Cooking frequency	Almost never; less than once a week; 1–2 times a week; 3–4 times a week; almost every day.
Experience with oven-type appliances	Use often; use sometimes; used once or twice; rarely or never used. Examples given: oven, lightwave oven, steam oven, air-fryer oven, and similar appliances.
Tendency to try new dishes or recipes	Seven-point agreement scale.
Interest in smart-appliance features	Seven-point agreement scale.

B.2. Survey Section 1: display and cooking-status visualization

Respondents saw the front-view comparison (Figure 16) and the open-door comparison (Figure 17). The accompanying explanation stated that A makes cooking status and next actions directly visible through a wide display and handle guidance, while B keeps a simpler screen and a more familiar physical structure. The forced-choice matrix used the options A, B, and Similar / Not sure, with the items listed in Table 7; the section closed with an overall preference for Survey Section 1 and an open-ended reason.



Figure 16. Survey Section 1 front-view stimulus, as shown in the form.



Figure 17. Survey Section 1 open-door and handle stimulus, as shown in the form.

Table 7. Survey Section 1 forced-choice items. Each row used A, B, and Similar / Not sure.

Item wording

Easy to understand the current cooking status.

Easy to check temperature and time.

Clear next action.

Easy for first-time users to learn.

Convenient to use in an actual kitchen.

Feels premium.

B.3. Survey Section 2: cooking support and automation

Respondents saw the cooking-support comparison (Figure 18). The explanation stated that A provides a Chef Persona with voice and visual feedback, recommendations, coaching, and progress guidance, while B provides automatic cooking for selected menus and leaves most decisions to the user. Table 8 lists the items.



Figure 18. Survey Section 2 cooking-support stimulus, as shown in the form.

Table 8. Survey Section 2 items.

Item type	Wording and response options
Forced-choice matrix	Each row used A, B, and Similar / Not sure: reduces uncertainty before starting to cook; reduces the burden of choosing mode, temperature, and time; increases confidence to try new recipes; helps reduce mistakes or failures during cooking; seems likely to be used regularly; feels like a smarter cooking experience overall.
Multiple selection	Which elements of A seem especially useful? Options: AI Agent recommendations; Chef persona coaching; voice / microphone interaction; automatic cooking-setting suggestions; next-step guidance; none of the above seems especially useful.
Overall preference	Overall, which direction do you prefer in Survey Section 2? Options: A; B; both are similar; not sure.
Open-ended reason	Please briefly explain why you chose A or B in Survey Section 2.

B.4. Final evaluation

With both sections in view, respondents completed the final items in Table 9.

Table 9. Final-evaluation items.

Question type	Wording and response options
Final forced-choice matrix	Overall, which direction feels better for each item? Rows: overall user experience; understanding the cooking process; reducing anxiety about cooking failure; likelihood of trying new recipes; real purchase consideration. Columns: A; B; Similar / Not sure.
Purchase-intent rating, A	If a product based on A were released, how likely would you be to consider purchasing it? Eleven-point scale, 0–10.
Purchase-intent rating, B	If a product based on B were released, how likely would you be to consider purchasing it? Eleven-point scale, 0–10.
Final forced choice, reason	If you had to choose one of the two directions, which would you choose, and why?
Additional comments	Please share any additional comments about the survey or the design direction.

C. Original Visual Materials

The main text uses figures redrawn for print legibility. This appendix preserves the originals behind them: the affinity board as built, and the per-task flow charts extracted from the midterm analysis slides and arranged for reading.

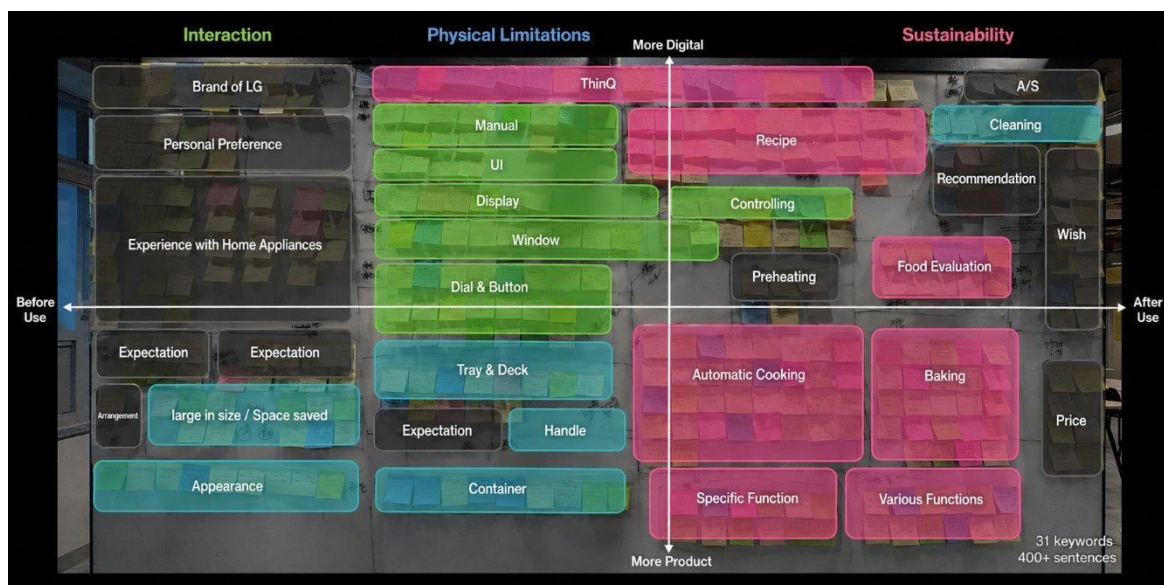


Figure 19. Original hybrid affinity diagram from User Study 1, as built on the project board.

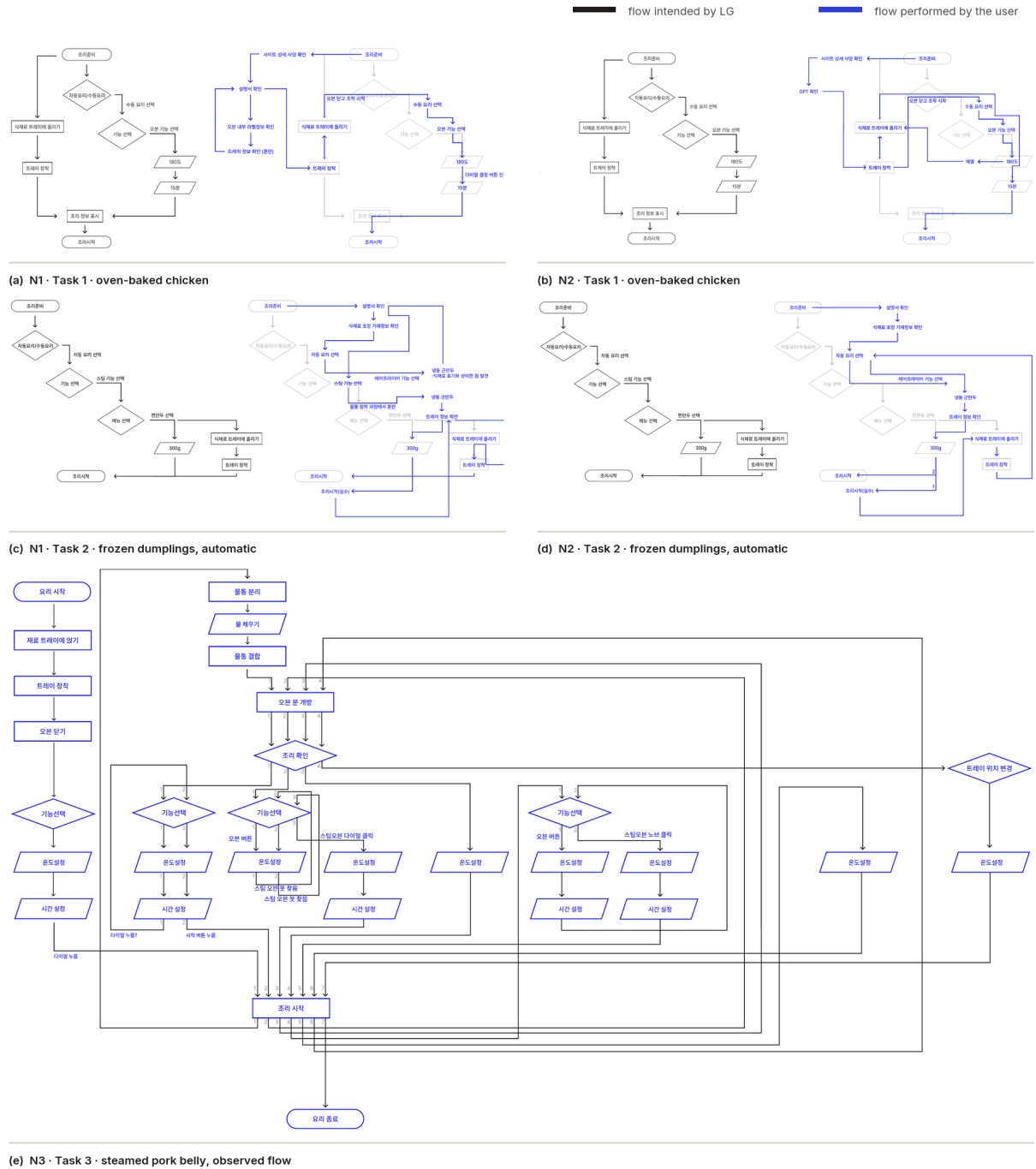


Figure 20. Intended-versus-observed flow charts from User Study 1, extracted from the midterm analysis slides and arranged by participant and task: (a, b) Task 1, oven-baked chicken; (c, d) Task 2, frozen dumplings with automatic cooking; (e) N3's steamed pork belly task. Black lines mark the flow intended by LG; blue lines mark the flow each participant performed.

D. Comparative Evaluation

D.1. Complete counts in instrument order

Table 10. All forced-choice items and section-level preferences, in instrument order. The section-overall items offered four options (A, B, Both are similar, Not sure); the two non-A/B options are pooled in the third column.

Item	A	Similar / not sure	B	n
<i>Survey Section 1: display and cooking-status visualization</i>				
Easy to understand the current cooking status	28	3	4	35
Easy to check temperature and time	23	5	7	35
Clear next action	17	6	12	35
Easy for first-time users to learn	9	3	23	35
Convenient to use in an actual kitchen	14	7	14	35
Feels premium	24	6	5	35
Overall preference within Survey Section 1	20	1	14	35
<i>Survey Section 2: cooking support and automation</i>				
Reduces uncertainty before starting to cook	24	7	5	36
Reduces the burden of choosing mode, temperature, and time	17	8	11	36
Increases confidence to try new recipes	21	10	5	36
Helps reduce mistakes or failures during cooking	18	10	8	36
Seems likely to be used regularly	11	3	22	36
Feels like a smarter cooking experience overall	20	9	7	36
Overall preference within Survey Section 2	14	3	19	36
<i>Final evaluation</i>				
Overall user experience	17	3	16	36
Understanding the cooking process	25	3	8	36
Reducing anxiety about cooking failure	22	8	6	36
Likelihood of trying new recipes	21	10	5	36
Real purchase consideration	14	4	18	36

D.2. Premium-experience index

For each respondent with a complete grid ($n = 35$), the index counts how many of the eleven premium-experience attributes in Table 3 were answered A, Similar / Not sure, or B. Means are 6.74 for the developed direction, 2.20 for Similar / Not sure, and 2.06 for the comparison design; Figure 12 plots these to one decimal. At the respondent level, 29 of 35 chose the developed direction on more premium attributes than the comparison design, five chose the comparison design on more, and one tied. Treating each non-tied respondent as a binary outcome, the exact two-sided sign test gives $p = .00004$ against an even split; the premium-experience preference is carried by individual respondents, not by a few extreme grids.

D.3. Purchase-intent ratings

Each respondent rated purchase intent for both directions on the 0–10 scale ($n = 36$). The developed direction received a mean of 5.33 and a median of 6; the comparison design a mean of 5.47 and a median of 5. Paired within respondents, 16 rated the developed direction higher, 15 the comparison design, and five gave equal ratings; a two-sided sign test does not distinguish the two ($p = 1.0$). The categorical purchase-consideration lean toward the comparison design in Table 3 therefore reflects a forced choice between a familiar and a novel direction, not a difference in stated willingness to consider purchase.